

Experiment 06: Design a Fuzzy Logic Controller

Learning Objective: To design and understand a fuzzy logic-based controller for automatic gear shifting in a Dual Clutch Transmission (DSG) system using fuzzy inference principles.

Tools: Python

Theory:

Fuzzy Logic is a form of reasoning based on approximate values rather than fixed and exact inputs. It is derived from the concept of Fuzzy Logic, which allows partial membership instead of binary true/false values.

A fuzzy logic controller (FLC) is widely used in real-world systems where precise mathematical models are difficult to develop. It mimics human decision-making by using linguistic variables such as low, medium, and high.

In automotive systems, fuzzy controllers are used to enhance driving comfort, efficiency, and performance. One such application is in the Dual Clutch Transmission (DSG), commonly used in vehicles manufactured by Volkswagen. DSG systems use two clutches to enable fast and seamless gear shifts.

The fuzzy controller in DSG systems determines optimal gear shifting based on input parameters such as vehicle speed and throttle position. Instead of fixed thresholds, fuzzy logic allows smooth transitions between gears, improving both performance and fuel efficiency.

Working Principle of Fuzzy Controller: The fuzzy logic controller operates in four main stages:

1. Fuzzification: The crisp input values (e.g., speed and throttle) are converted into fuzzy values using membership functions. Each input is assigned degrees of membership in predefined fuzzy sets.

2. Rule Base: A set of IF-THEN rules is defined based on human driving behavior. Example:

- IF speed is LOW AND throttle is HIGH → Downshift
- IF speed is HIGH AND throttle is LOW → Upshift

3. Inference Engine: The inference engine evaluates all the rules and determines the degree to which each rule is applicable. It combines the results to produce fuzzy outputs.

4. Defuzzification: The fuzzy output is converted into a crisp value, which represents the final gear decision (upshift, downshift, or hold).

Fuzzy Variables Used:

Input Variables:

1. **Speed (0–120 km/h):**
 - Low

- Medium
 - High
2. **Throttle Position (0–100%):**
- Low
 - Medium
 - High

Output Variable:

1. **Gear Action:**
- Downshift
 - Hold
 - Upshift

Membership Functions: Triangular membership functions are used for simplicity and efficiency.

- Speed is divided into Low, Medium, High ranges
- Throttle is divided into Low, Medium, High ranges
- Output (Gear Action) is mapped between -1 and +1

Overlapping of membership functions ensures smooth transitions between gear states.

Fuzzy Rule Base: The following rules are used for decision-making:

1. IF Speed is LOW AND Throttle is HIGH → Downshift
2. IF Speed is HIGH AND Throttle is LOW → Upshift
3. IF Speed is MEDIUM AND Throttle is MEDIUM → Hold
4. IF Speed is HIGH AND Throttle is HIGH → Hold
5. IF Speed is LOW AND Throttle is LOW → Hold
6. IF Speed is MEDIUM AND Throttle is HIGH → Downshift
7. IF Speed is HIGH AND Throttle is MEDIUM → Upshift

Algorithm:

Step 1: Define input variables (Speed, Throttle) and output variable (Gear Action).

Step 2: Define membership functions for all variables.

Step 3: Convert crisp inputs into fuzzy values (Fuzzification).

Step 4: Apply fuzzy rules using inference engine.

Step 5: Aggregate results of all rules.

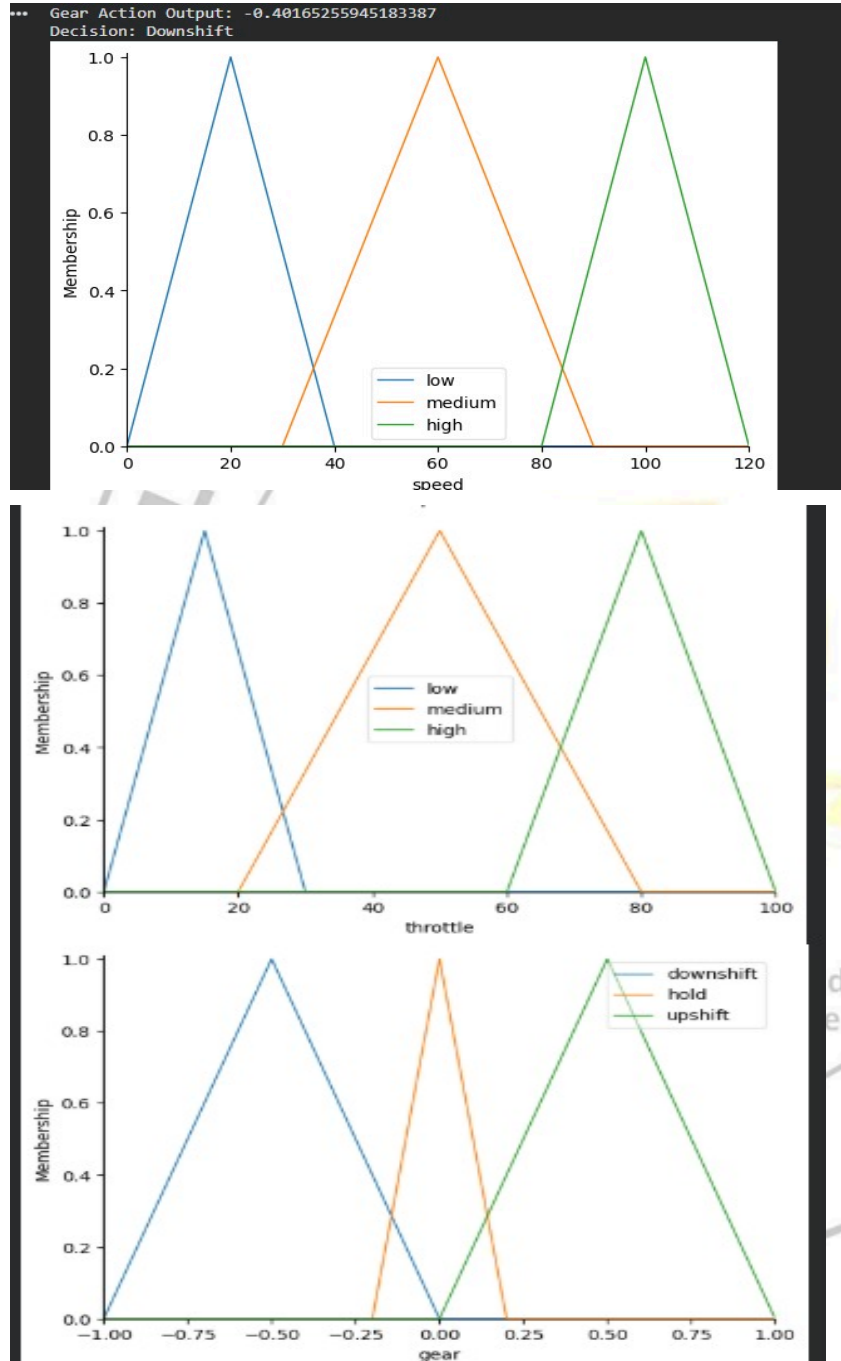
Step 6: Convert fuzzy output into crisp output (Defuzzification).

Step 7: Display the gear decision (Upshift / Downshift / Hold).

Implementation Code:

<pre>!pip install scikit-fuzzy import numpy as np import skfuzzy as fuzz from skfuzzy import control as ctrl # 1. Define Inputs & Output # Speed (0–120 km/h) speed = ctrl.Antecedent(np.arange(0, 121, 1), 'speed') # Throttle (0–100 %) throttle = ctrl.Antecedent(np.arange(0, 101, 1), 'throttle') # Gear action (-1 to 1) gear = ctrl.Consequent(np.arange(-1, 1.1, 0.1), 'gear') # 2. Membership Functions # Speed speed['low'] = fuzz.trimf(speed.universe, [0, 20, 40]) speed['medium'] = fuzz.trimf(speed.universe, [30, 60, 90]) speed['high'] = fuzz.trimf(speed.universe, [80, 100, 120]) # Throttle throttle['low'] = fuzz.trimf(throttle.universe, [0, 15, 30]) throttle['medium'] = fuzz.trimf(throttle.universe, [20, 50, 80]) throttle['high'] = fuzz.trimf(throttle.universe, [60, 80, 100]) # Gear Action gear['downshift'] = fuzz.trimf(gear.universe, [-1, -0.5, 0]) gear['hold'] = fuzz.trimf(gear.universe, [-0.2, 0, 0.2]) gear['upshift'] = fuzz.trimf(gear.universe, [0, 0.5, 1]) # 3. Define Rules rule1 = ctrl.Rule(speed['low'] & throttle['high'], gear['downshift']) rule2 = ctrl.Rule(speed['high'] & throttle['low'], gear['upshift']) rule3 = ctrl.Rule(speed['medium'] & throttle['medium'], gear['hold']) rule4 = ctrl.Rule(speed['high'] & throttle['high'], gear['hold']) rule5 = ctrl.Rule(speed['low'] & throttle['low'], gear['hold']) rule6 = ctrl.Rule(speed['medium'] & throttle['high'], gear['downshift']) rule7 = ctrl.Rule(speed['high'] & throttle['medium'], gear['upshift']) # 4. Control System gear_ctrl = ctrl.ControlSystem([rule1, rule2, rule3, rule4, rule5, rule6, rule7]) gear_sim = ctrl.ControlSystemSimulation(gear_ctrl) # 5. Test the System # Example inputs gear_sim.input['speed'] = 50 gear_sim.input['throttle'] = 70 # Compute output gear_sim.compute() print("Gear Action Output:", gear_sim.output['gear']) # 6. Interpret Output output = gear_sim.output['gear'] if output < -0.2: print("Decision: Downshift") elif output > 0.2: print("Decision: Upshift") else: print("Decision: Hold Gear") # 7. (Optional) Visualize speed.view() throttle.view() gear.view()</pre>	
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Output:



Result and Discussion:

Learning Outcomes: Students should have the ability to

LO 1: Understand the concept of fuzzy logic and its application in control systems.

LO 2: Design a fuzzy logic controller using membership functions and rule base.

LO 3: Analyze how fuzzy logic improves decision-making in automotive systems.

Course Outcomes: Understand and apply fuzzy logic controller concepts in real-life applications such as automotive transmission systems.

Conclusion:

Viva Questions:

Q.1 What is a fuzzy logic controller and how does it differ from a conventional controller?

Q.2 Why is fuzzy logic suitable for controlling a DSG transmission system?

Q.3 What is the role of membership functions in a fuzzy inference system?

For Faculty Use

Sr. No.	Parameters	Weightage	Category	Performance				
				Excellent	Very Good	Good	Average	Below Average
1	Technical Knowledge (Preparedness and Execution)	/30	Prerequisites	12	10	8	6	4
			Skills in performing experiment	12	10	8	6	4
			Learning	6	5	4	3	2
2	Technical Documentation (Developing skills for Journal Writing and Maintenance of Lab notebook)	/20	Format	4	3	2	1	0
			Contents as per format	8	6	4	2	0
			Quality	8	6	4	2	0
3	Level of Interaction (Developing Expression Power)	/10	Level of understanding	4	3	2	1	0
			Questions & Answers	4	3	2	1	0
			Application	2	2	1	1	0
4	Behavioral (Attitude towards Learning)	/20	Attitude	4	3	2	1	0
			Regularity	8	6	4	2	0
			Team Work/Group Activity	8	6	4	2	0
5	Compliance (Attainment of Objectives/Outcomes)	/20	Course Attainment	12	10	8	6	4
			Learning objective attainment	8	6	4	2	0
	TOTAL	/100						